

WEEK-3

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COMPARISON OPERATOR \$gt

OBJECTIVE: Find books having a rating greater than 4.

The screenshot shows the MongoDB Compass interface. At the top, the organization is 'SIMRAN JANGID's Or...' and the project is 'Project 0'. The database is 'book_metadata'. The query editor shows a query: `{ "reviews.rating": { "$gt": 4 } }`. The results are displayed in a table with 5 documents.

	ObjectId	book_id Int32	title String
1	jectId('6a8f19a318e58ee...	101	"The Martian"
2	jectId('6a8f19c718e58ee...	102	"Dune"
3	jectId('6a8f19db18e58ee...	103	"Clean Code"
4	jectId('6a8f19ed18e58ee...	104	"Artificial Intell"
5	jectId('6a8f19fe18e58ee...	105	"Data Structures"

USING \$IN

Find books containing reviews with ratings 4 or 5.

ORGANIZATION
SIMRAN JANGID's Or...

PROJECT
Project 0

transactions

bookflow

book_metadata

+

Cluster0 > bookflow > book_metadata

View monitoring

Visualize Yo

Documents 5

Aggregations

Schema

Indexes 1

Validation

🕒

{

"reviews.rating": {

"\$in": [4, 5]

}

}

✦

Explain

Reset

Find

Opti

+

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💡

25

1 – 5 of 5

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⋮

{

🏠 book_metadata

		book_id Int32	title String	author Str	
1	58ee...	101	"The Martian"	"Andy Weir"	✎ 🗑
2	58ee...	102	"Dune"	"Frank Her"	✎ 🗑
3	58ee...	103	"Clean Code"	"Robert C."	✎ 🗑
4	58ee...	104	"Artificial Intelligence ..."	"John Smit"	✎ 🗑
5	58ee...	105	"Data Structures"	"Mark Alle"	✎ 🗑

REGEX QUERY

Find books where the format contains the word Digital

The screenshot shows the Data Explorer interface. On the left, the 'CLUSTERS' sidebar lists 'Cluster0' with sub-items: 'admin', 'bankdb', 'bookflow', 'books', 'book_metadata', 'local', and 'sample_mflix'. The 'book_metadata' cluster is selected. The main panel shows the 'Documents' tab for 'Cluster0 > bookflow > book_metadata'. A query is entered in the search bar:

```
{ "format": { "$regex": "Digital", "$options": "i" } }
```

. Below the query, there are buttons for 'ADD DATA', 'BULK', and 'EXPORT CODE'. A table of results is displayed below the query:

	title String	author String	published_year Int32	category String	format String
1	"The Martian"	"Andy Weir"	2021	"Science Fiction"	"Digital PDF"
2	"Dune"	"Frank Herbert"	2022	"Science Fiction"	"Digital EPUB"
3	"Artificial Intelligence ..."	"John Smith"	2023	"Artificial Intelligence"	"Digital PDF"
4	"Data Structures"	"Mark Allen"	2024	"Computer Science"	"Digital PDF"

AGGREGATION PIPELINE

OBJECTIVE: Generate a Top Rated report.

STEP 1: \$MATCH

The screenshot shows the Data Explorer interface with the 'Aggregations' tab selected for 'Cluster0 > bookflow > book_metadata'. A new aggregation stage named '\$match' is added to the pipeline. The query editor shows the following pipeline stages:

```
1 /**
2  * query: The query in MQL.
3  */
4 {
5   "published_year": {
6     "$gt": 2020
7   }
8 }
```

On the right, the 'Output preview after \$match stage (Sample of 4 documents)' is shown:

```
{ "_id": ObjectId('6a8f19a...'),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fic",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101,
  "reviews": Array (3) }
```

STEP 2: Add \$UNWIND, \$unwind, \$group and \$sort

The screenshot shows the MongoDB Atlas Data Explorer interface. The left sidebar displays the 'Data Explorer' section with 'My Queries' and 'Data Modeling' options. The 'CLUSTERS (1)' section shows a tree view with 'Cluster0' expanded, containing 'admin', 'bankdb', 'bookflow', 'book_metadata', 'books', 'local', and 'sample_mflix'. The main panel shows the 'book_metadata' collection with a pipeline editor. The pipeline stages are: '\$match', '\$unwind', '\$group', and '\$sort'. The '\$match' stage has a query: 'The query in MQL.'. The '\$unwind' stage has a path: 'published_year: { \$gt: 2020 }'. The '\$group' stage has a path: 'published_year: { \$gt: 2020 }'. The '\$sort' stage has a path: 'published_year: { \$gt: 2020 }'. The 'PIPELINE OUTPUT PREVIEW' section shows a sample of 4 documents:

```
{ "_id": "The Martian", "average_rating": 4.666666666666667, "total_reviews": 3 }
```

```
{ "_id": "Dune", "average_rating": 4.666666666666667, "total_reviews": 3 }
```

```
{ "_id": "Artificial Intelligence ...", "average_rating": 4.666666666666667, "total_reviews": 3 }
```

CREATE TEXT INDEX

OBJECTIVE: Create a text index on the comments field inside reviews.

The screenshot shows the MongoDB Atlas Data Explorer interface. The left sidebar displays the 'Data Explorer' section with 'My Queries' and 'Data Modeling' options. The 'CLUSTERS (1)' section shows a tree view with 'Cluster0' expanded, containing 'admin', 'bankdb', 'bookflow', 'book_metadata', 'books', 'local', and 'sample_mflix'. The main panel shows the 'book_metadata' collection with a pipeline editor. The pipeline stages are: '\$match', '\$unwind', '\$group', and '\$sort'. The '\$match' stage has a query: 'The query in MQL.'. The '\$unwind' stage has a path: 'published_year: { \$gt: 2020 }'. The '\$group' stage has a path: 'published_year: { \$gt: 2020 }'. The '\$sort' stage has a path: 'published_year: { \$gt: 2020 }'. The 'PIPELINE OUTPUT PREVIEW' section shows a sample of 4 documents:

```
{ "_id": "The Martian", "average_rating": 4.666666666666667, "total_reviews": 3 }
```

```
{ "_id": "Dune", "average_rating": 4.666666666666667, "total_reviews": 3 }
```

```
{ "_id": "Artificial Intelligence ...", "average_rating": 4.666666666666667, "total_reviews": 3 }
```

SEARCH FOR A SPECIFIC KEYWORD

Search for: science fiction

The screenshot shows the MongoDB Atlas interface. On the left, the 'Explorer' sidebar lists the organization 'SIMRAN JANGID's Or...' and project 'Project 0'. The 'book_metadata' collection is selected. The main panel shows the 'Documents' tab with a search query: `{ "$text": { "$search": "\"science fiction\"" } }`. The query is executed, and the results are displayed in a table with columns: 'author', 'published_year', and 'category'. The results show two documents: one by 'Frank Herbert' published in 2022, and another by 'Andy Weir' published in 2021, both categorized as 'Science Fiction'.

	author String	published_year Int32	category String
1	"Frank Herbert"	2022	"Science Fiction"
2	"Andy Weir"	2021	"Science Fiction"

USE explain()

OBJECTIVE: Check whether MongoDB is using an index or performing a collection scan.

The screenshot shows the 'Explain Plan' tab in MongoDB Atlas. It displays the execution plan for the search query. The plan indicates that the query is using a text index. The 'stage' is 'FETCH', and the 'inputStage' is 'IXSCAN'. The 'executionTimeMillisEstimate' is 0, and the 'works' are 6. The 'advanced' state is 5, and the 'needTime' is 0. The 'needYield' is 0, and the 'saveState' is 0. The 'restoreState' is 0, and the 'isEOF' is 1. The 'docsExamined' is 5, and the 'alreadyHasObj' is 0. The 'inputStage' is 'IXSCAN', and the 'nReturned' is 5. The 'executionTimeMillisEstimate' is 0, and the 'works' are 6. The 'advanced' state is 5, and the 'needTime' is 0. The 'needYield' is 0, and the 'saveState' is 0. The 'restoreState' is 0, and the 'isEOF' is 1. The 'docsExamined' is 5, and the 'alreadyHasObj' is 0.